MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.3 Worksheet

1. Which of the following expressions are defined? Which are meaningless? Explain.

a) $(\mathbf{a}\cdot\mathbf{b})\cdot\mathbf{c}$

b) $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

c) $\|\mathbf{a}\|(\mathbf{b}\cdot\mathbf{c})$

d) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$

e) $\mathbf{a}\cdot\mathbf{b}+\mathbf{c}$

f) $\|\mathbf{a}\|\cdot(\mathbf{b}+\mathbf{c})$

2. Find $\mathbf{a}\cdot\mathbf{b}$ for each of the following sets of vectors and determine the angle between each of these vectors. Round angles to 2 decimal places and express these angles in radians.

a) $\mathbf{a}=\langle -2,3\rangle$ & $\mathbf{b}=\langle 0.7,1.2\rangle$

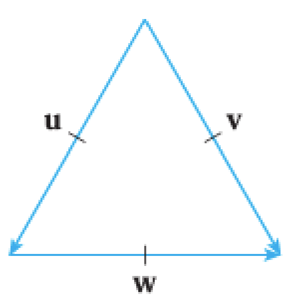
b) $\mathbf{a}=\langle 6,-2,3\rangle$ & $\mathbf{b}=\langle 2,5,-1\rangle$

c) $\mathbf{a}=2\mathbf{i}+\mathbf{j}$ & $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$

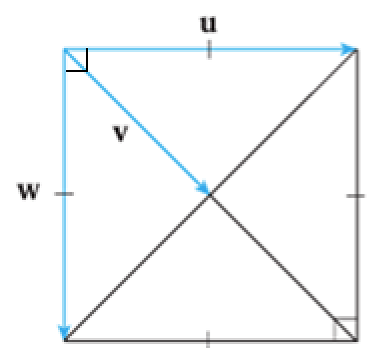
d) $\mathbf{a}=4\mathbf{i}-3\mathbf{j}+\mathbf{k}$ & $\mathbf{b}=2\mathbf{i}-\mathbf{k}$

3. If $\mathbf{u}$ is a unit vector, find $\mathbf{u}\cdot\mathbf{v}$ and $\mathbf{u}\cdot\mathbf{w}$.

a)



b)



4. Determine the angle between the vector $\mathbf{u} = \langle 1, 2, 3 \rangle$ and each of the coordinate axes. Round angles to 2 decimal places and express these angles in radians.

5. Determine if the given vectors are orthogonal, parallel, or neither.

a) $\mathbf{a} = \langle -5, 3, 7 \rangle$ & $\mathbf{b} = \langle 6, -8, 2 \rangle$

b) $\mathbf{a} = \langle 4, 6 \rangle$ & $\mathbf{b} = \langle -3, 2 \rangle$

c) $\mathbf{a} = -\mathbf{i} + 2\mathbf{j} + 5\mathbf{k}$ & $\mathbf{b} = 3\mathbf{i} + 4\mathbf{j} - \mathbf{k}$

d) $\mathbf{a} = 2\mathbf{i} + 6\mathbf{j} - 4\mathbf{k}$ & $\mathbf{b} = -3\mathbf{i} - 9\mathbf{j} + 6\mathbf{k}$

6. Find the component and projection of **b** onto **a**. In the 2-dimensional cases include a sketch of these vectors and the projection (sketch all vectors starting from the origin).

a) $\mathbf{a} = \langle -5, 12 \rangle$ & $\mathbf{b} = \langle 4, 6 \rangle$

b) $\mathbf{a} = \langle 1, 4 \rangle$ & $\mathbf{b} = \langle 2, 3 \rangle$

c) $\mathbf{a} = \langle -2, 3. -6 \rangle$ & $\mathbf{b} = \langle 5, -1, 4 \rangle$

d) $\mathbf{a} = 2\mathbf{i} - \mathbf{j} + 4\mathbf{k}$ & $\mathbf{b} = \mathbf{j} + \frac{1}{2}\mathbf{k}$

7. Suppose that **a** and **b** are nonzero vectors.

a) Under what circumstances is $comp_{\mathbf{a}}(\mathbf{b}) = comp_{\mathbf{b}}(\mathbf{a})$.

b) Under what circumstances is $proj_{\mathbf{a}}(\mathbf{b}) = proj_{\mathbf{b}}(\mathbf{a})$.

8. If $\mathbf{c} = \|\mathbf{a}\|\mathbf{b} + \|\mathbf{b}\|\mathbf{a}$, where **a**, **b**, and **c** are all nonzero vectors, show that **c** bisects the angle between **a** and **b.**

9. Show that if $\mathbf{u} + \mathbf{v}$ and $\mathbf{u} - \mathbf{v}$ are orthogonal, then the vectors **u** and **v** must have the same length.

10. If $\mathbf{r} = \langle x, y, z \rangle$, $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$, & $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$, show that the vector equation $(\mathbf{r} - \mathbf{a}) \cdot (\mathbf{r} - \mathbf{b}) = 0$ represents a sphere, and finds its center and radius.